



BERLIN SUMMER LOCK-IN 2026 · CORTEA FRAUD-HUNT TRACK

fraudmind

Fraud hunting with proof attached.

No number without a source. Every finding cites document and row.

Engine decides numbers

AI writes language

Auditor decides verdicts

Machines surface evidence. Auditors keep judgment.

01 · RULES

Deterministic engine

Links ledgers, subledgers, assets and approvals into transaction chains; runs 22 check families; cites every figure to file and row.

Thresholds, lock dates and management IDs are parsed from the dossier's own audit-planning workpaper — zero hardcoded identifiers.

02 · LANGUAGE + GRAPH

Constrained AI layer

OpenAI writes auditor-ready explanations; a Cogne knowledge graph independently ranks fraud candidates from relationship facts.

Every numeric token in AI text is validated against engine output — the model can explain a number, never invent one.

03 · VERDICTS

Human review

Confirm fraud, request follow-up, or dismiss — with reviewer notes. Verdicts and citations become the formal verification report.

Two-tier findings (FLAGGED vs NEEDS_REVIEW) keep judgment calls out of the accusation tier.

A million ledger lines become a review queue — not a black box.

1

Ingest the dossier

GDPdU exports, companion documents and subledgers parsed and linked by document number. Rename-tolerant: fuzzy file resolution with year-token guards.

2

Run traceable tests

Control definitions and thresholds read from the dossier's own planning workpaper; base rates computed from the data itself.

3

Inspect the evidence

Every case shows tier, confidence, amount and clickable file:row citations that open the actual source record.

4

Record the verdict

Auditor decisions and notes become a PDF / Markdown verification report — and fraud nodes in the knowledge graph.

LIVE DEMO

fraudmind.netlify.app

It held up on an unseen 1.08M-line dossier.

1,083,723

GL lines analyzed on the previously unseen final dossier

54

ranked findings · 20 flagged · 33 needs review · 1 info

~40 sec

end-to-end runtime · same pipeline, unchanged

Top finding: €6,000,000 journal posted “GEBUCHT OHNE FREIGABE”

Management user BSP-U09 · 30.12.2025 at 22:47 · no release record · six times materiality. Corroborated by four quarter-end €1.1M self-approved journals (BSP-U02) and €3.59M of post-lock-date entries.

Human review confirmed **6 frauds · €13,987,881** — the knowledge graph independently ranks the same candidates.

Pilot fraudmind on one real dossier.

Measure what auditors care about:

- Review time per dossier — hours, not weeks
- Evidence coverage — every claim traceable to file : row
- False-positive rate — the two-tier system's core promise

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SOURCE

github.com/kaiser-data/fraudmind

Calibrated on ground truth, guarded against overfitting.

Scored against a known answer key

Practice dossier: 4 of 4 planted frauds found, zero seeded decoys wrongly accused. Every key amount re-derived from raw source files — net vs gross reconciled to the cent.

EVIDENCE: PRACTICE KEY COMPARISON

Formal calibration loop

Each engine-vs-key delta classified (TP / corroboration / miss / FP); misses diagnosed at the invariant level. Every fix passes a generalization gate: zero identifiers from any dossier, byte-identical regression rerun, year-shifted rename simulation.

EVIDENCE: GT CALIBRATION HANDOFF

Base-rate self-calibration

Baselines computed from the dossier itself (practice: 28.5% weekend postings; final: 11% night). A pattern is flagged only if rarer than 5% — and the baseline is emitted as an INFO finding, so the calibration is auditable.

EVIDENCE: JET_TIME_PROFILE FINDING

Two-tier vs the false-positive penalty

FLAGGED is reserved for deterministic violations. When 88 of 322 credit-limit rows mismatched, the engine recognized a comparison-basis issue and raised ONE review item — not 88 accusations.

EVIDENCE: FINAL-RUN FINDING F029

Built in one day — because the agent carries written policies.

Learned skills as institutional memory

audit-detectors: numbered anti-overfitting policies distilled from scored evidence (1,170 raw findings → 18, zero frauds lost). investigating-financial-anomalies: hypothesis families, benign-explanation testing, calibrated confidence language.

Ingest — stdlib + openpyxl

GDPdU tables via index.xml with content-based fallbacks; cross-document transaction chains; document prefixes and user-ID patterns derived from the data. 1.08M GL rows in ~27 seconds.

Detection — 22 deterministic families

Self-approval & SoD, posted-without-release, lock-date breaches, cut-off both directions, rare accounts, credit limits, tie-outs, sequence gaps — all parameterized by the dossier's own planning worksheet at runtime.

Constrained AI + serving

OpenAI explanation layer with per-token numeric validation; Cognee Cloud knowledge graph (6 documents incl. human verdict nodes) for independent ranking and live Q&A; stdlib server keeps keys server-side; React review console with PDF report.